

Shared architectural scope
Two fast-multiply Ibex harts · GEMM · SFPU · autonomous memory/transfer control
Full logical capacities, numerical formats and coherent local storage

FPGA implementation

PYNQ-Z1 / Zynq-7000 programmable logic
LUTs, registers and block RAM
Qualified system clock: 91 MHz

↕ AXI

Zynq processing system and shell
ARM/Linux provisioning and request control
SmartConnect, clock/reset
and DDR controller

↕

Board DDR memory
Full model and working arena

ASIC implementation

Sky130 standard cells
+ characterized SRAM
ASIC-specific memory and clock bindings
System target: 100 MHz; closure pending

↕ control + memory

ASIC platform boundary
SPI loader / UART, clock/reset and pads
External-memory
controller/interface qualification

↕

External memory required
Not replaced by a
micro-weight demonstration

FPGA performance is measured on the board. ASIC area/timing are implementation-tool results;
no fabricated ASIC, measured ASIC token rate or fabrication-ready claim is implied.